

ECON 311

Lecture 11: Inflation and Introduction to SR Macroeconomics

Growth Accounting

Inflation

Short-Run Macroeconomics

Solow and Romer

We combine Romer and Solow:

$$Y_t = F(K_t, L_t, A_t) = A_t K_t^{\frac{1}{3}} L_t^{\frac{2}{3}}$$

$$L_{A_t} = \bar{l} \bar{L}$$

$$L_{Y_t} = (1 - \bar{l}) \bar{L}$$

$$\underbrace{\frac{A_{t+1} - A_t}{A_t}}_{g_{A_t}} = \bar{z} L_{A_t}$$

$$\underbrace{\frac{K_{t+1} - K_t}{K_t}}_{g_{K_t}} = \bar{s} \frac{Y_t}{K_t} - \bar{d}$$

Solow and Romer

- In the long-run equilibrium (BGP), we need capital to grow at a constant rate, g_K , which means:

$$g_Y = g_K$$

- Using the above equation and the growth rate of TFP, we can solve for the long-run growth of Y_t :

$$g_Y = g_A + \frac{1}{3} g_k + \frac{2}{3} g_{(1-\bar{l})\bar{L}}$$

$$g_Y = \bar{z} \bar{l} \bar{L} + \frac{1}{3} g_Y + \frac{2}{3} 0$$

$$g_Y = g_y = \frac{3}{2} \bar{z} \bar{l} \bar{L} \geq 0$$

Growth Accounting

- Similar to the Production Function analysis, Growth Accounting disaggregates growth of output into the growth of inputs:

$$g_y = g_A + \frac{1}{3}(g_k - g_L) + \frac{2}{3}(g_{L_{Y_t}} - g_{L_t})$$

Growth Accounting for the United States

	1948–2014	1948–1973	1973–1995	1995–2007	2007–2014
Output per hour, Y/L	2.4	3.3	1.6	2.8	1.4
Contribution of K/L	0.9	1.0	0.8	1.1	0.6
Contribution of labor composition	0.2	0.2	0.2	0.2	0.3
Contribution of TFP, A	1.3	2.1	0.6	1.5	0.5

The table shows the average annual growth rate (in percent) for different variables.
Source: Bureau of Labor Statistics, *Multifactor Productivity Trends*.

Growth Accounting

Inflation

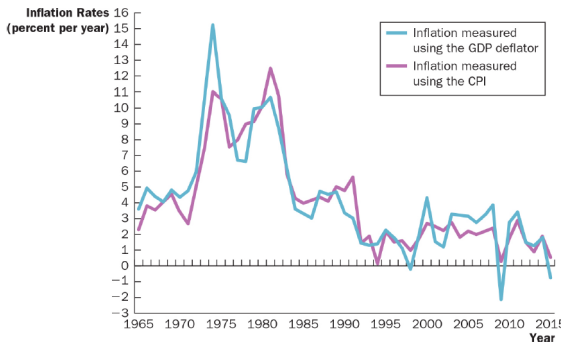
Short-Run Macroeconomics

Introduction

- Inflation rate is defined as the rate of change of prices

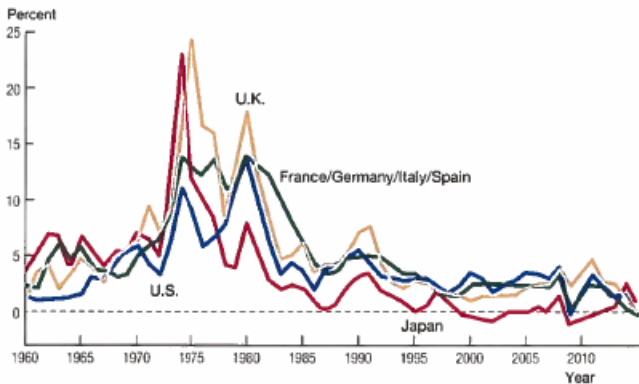
$$\pi_{t+1} = \frac{P_{t+1} - P_t}{P_t}$$

- Inflation in the Canadian economy:

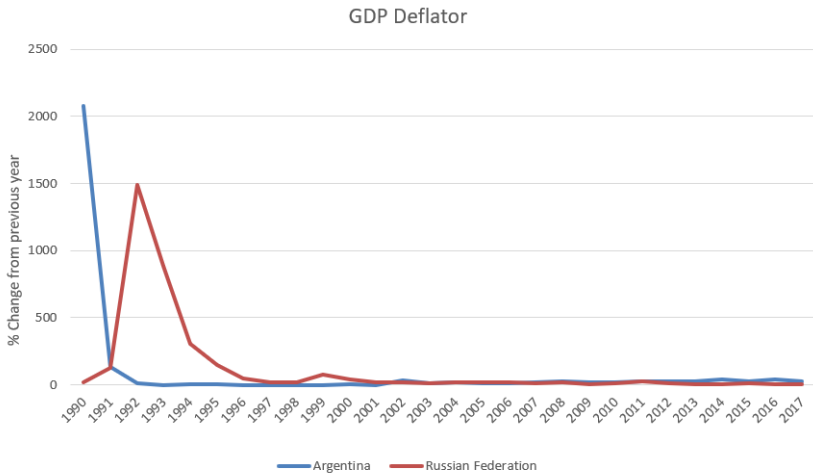


Introduction

Inflation Rates in Certain Rich Countries



Introduction



The Quantity Theory of Money

- The central component of the quantity theory of money is the quantity equation, which connects nominal GDP with the effective amount of liquidity in circulation:

$$M_t V_t = P_t Y_t$$

- ▶ M_t =Money Supply
 - ▶ V_t =Velocity (the rate at which money changes hands)
 - ▶ P_t =Price level
 - ▶ Y_t =Real GDP
- Effective money supply is equal to money demand

Prices and Inflation in the Quantity Theory

- Classical Dichotomy: real and nominal sides of the economy are separate (money is **neutral** in the long run).
- Prices in the quantity theory of money are given by:

$$P_t = \frac{M_t \bar{V}}{Y_t}$$

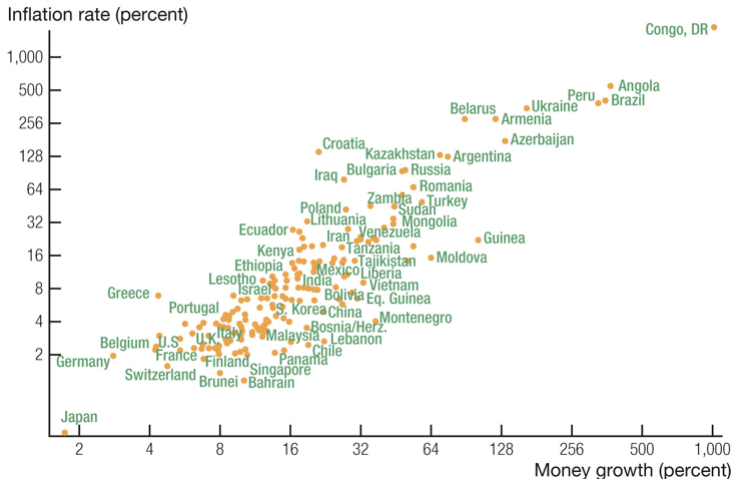
- ▶ Prices are given by how many goods the effective supply of money is chasing.
- Inflation is given by the following equation:

$$\Pi = g_M - g_Y$$

- ▶ If the classical dichotomy holds, we should observe that, on average and over the long-run, high inflation correlates with high levels of money supply growth.

Inflation in the Quantity Theory

Money Growth and Inflation around the World, 1990–2011



Risk of Inflation

- As we will see in the short-run analysis, monetary policy is an important tool to fight recessions (such as the COVID recession)
- A risk with loose monetary policy is the risk of increasing the inflation rate over the long-run
 - ▶ Costs associated with large levels of inflation are well understood.
 - ▶ Large debate over whether this risk has been overblown by economists in recent times (MMT)

Fisher Equation

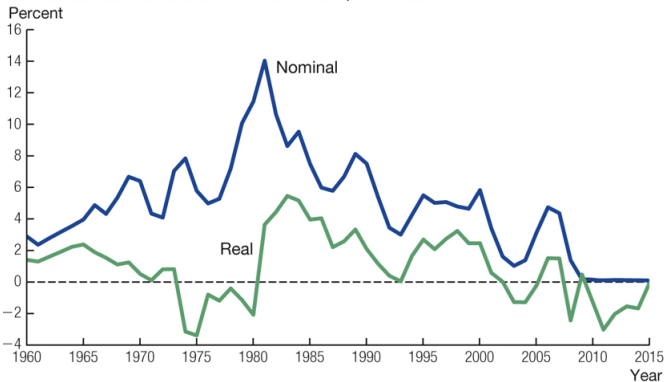
- The Fisher Equation allows us to convert nominal returns on investment into real returns:

$$i = R + \Pi \Rightarrow R = i - \Pi$$

- ▶ i : nominal interest rate
 - ▶ R : real interest rate
 - ▶ Π : inflation rate
- R depends on the MPK , which is usually stable in the short-run
 - ▶ Therefore, the nominal interest rate is usually high when inflation is high

Fisher Equation

Real and Nominal Interest Rates in the United States, 1960–2015



Growth Accounting

Inflation

Short-Run Macroeconomics

Short-Run Macroeconomics

- The long-run model is a guide to how the economy behaves on average
 - ▶ Determines potential output and long-run inflation
- Potential output
 - ▶ Amount of production if all inputs were utilized at their long-run sustainable levels
- At any given time, the economy is unlikely to exactly equal the long-run average
- The short-run model:
 - ▶ Determines current output and current inflation

Short-Run Macroeconomics

- Actual output is equal to the long-run trend plus short-run fluctuations:

$$\underbrace{\text{actual output}}_{Y_t} = \underbrace{\text{long-run trend}}_{\bar{Y}_t} + \underbrace{\text{short-run fluctuations}}_{\text{function of } \tilde{Y}_t}$$

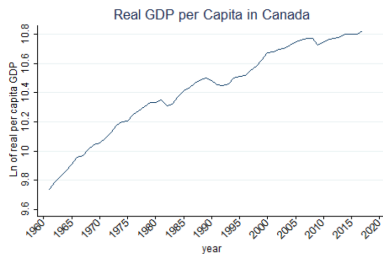
- ▶ Long-run trend: potential output
- ▶ Short-run fluctuations: deviations from $\bar{Y}_t$

Short-Run Macroeconomics

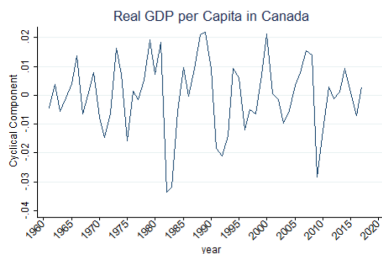
- “De-trended output” or short-run output:
 - ▶ The difference in actual and potential output, expressed as a percentage of potential output

$$\tilde{Y}_t = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}$$

Short-Run Macroeconomics



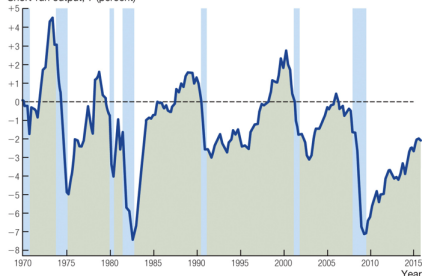
GDP per capita at 2007 constant prices
Source: Author's calculations from CANSIM data



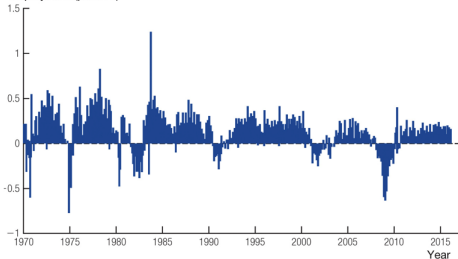
GDP per capita at 2007 constant prices
Source: Author's calculations from CANSIM data

Economics Fluctuations and Unemployment

U.S. Economic Fluctuations, 1970–2015
Short-run output, $\tilde{Y}$ (percent)



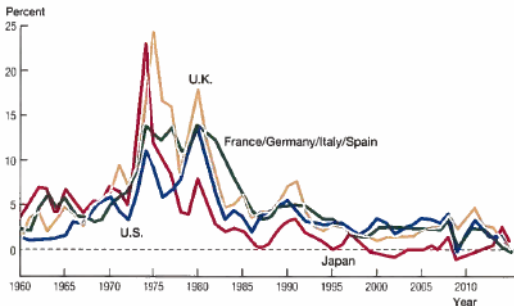
Changes in U.S. Employment, 1970–2015
Monthly change in employment (percent)



- Recessions are very costly:
 - ▶ E.g. during a typical recessions GDP per households is around \$12,000 below potential output
 - ▶ Unemployment has long-run negative effects on the population.

Inflation in the Short Run

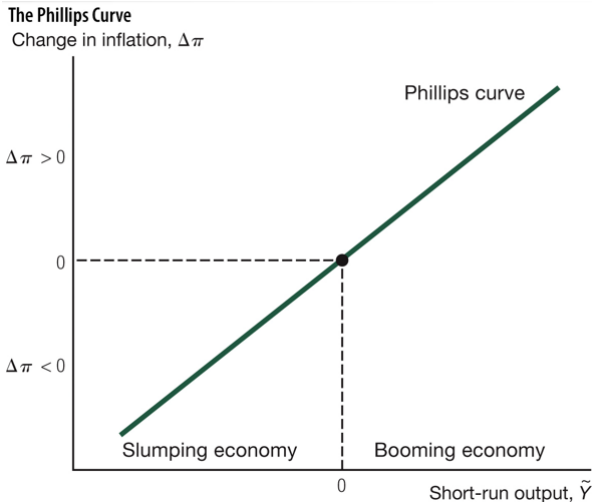
Inflation Rates in Certain Rich Countries



- Inflation increases during economic booms
- Inflation decreases during recessions

Output-Inflation Trade-off

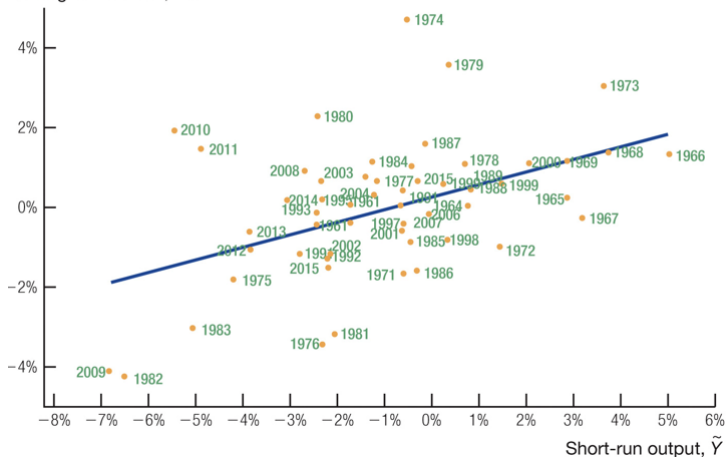
- The Phillips Curve denotes the output-inflation trade-off:



Phillips Curve in the US

Measuring the Phillips Curve, 1960–2015

Change in inflation, $\Delta\pi$



The Basis of the Short-Run Model

- The economy is constantly being hit by shocks that cause fluctuations in economic activity
 - ▶ Positive shocks increase economic activity beyond its natural rate, which results in inflation.
 - ▶ Negative shocks decrease economic activity below its natural rate, which results in recessions.
- Government through fiscal and monetary policy attempts to neutralize the effects of the shocks.
- This creates a constant cycle of boom and bust.
 - ▶ The duration and severity depends on the shock and government's reaction to the shock.

Economic Activity and Unemployment

- The Short-Run Model analyzes fluctuations in output and not in unemployment
- There is a direct connection between unemployment and output, known as Okun's Law:

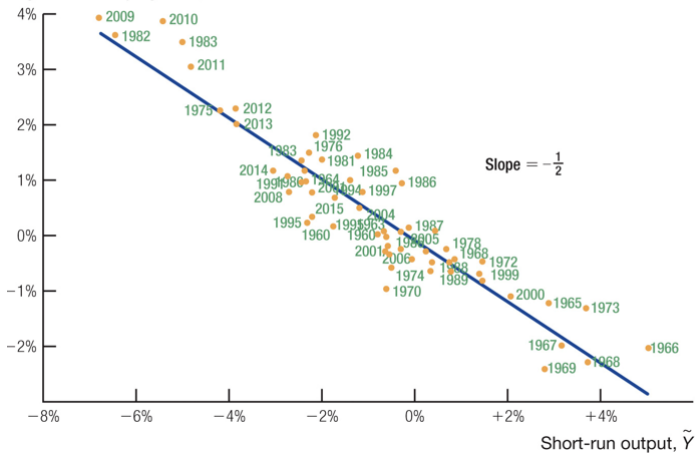
$$u - \bar{u} = -\frac{1}{2} \tilde{Y}$$

- ▶ u : current rate of unemployment
- ▶ $\bar{u}$: natural rate of unemployment
- ▶ $\tilde{Y}$: short-run output

Okun's Law in the US

Okun's Law for the U.S. Economy, 1960–2015

Cyclical unemployment, $u - \bar{u}$



Preview of the Short-Run Model

In the remainder of the semester:

1. We will carefully derive the IS-MP model and study the effects of shocks and government stabilization policy in this framework.
2. Afterward, we will study the connection between the IS-MP model and the Phillips curve.
3. Finally, we will put all the parts together and derive a model of the short-run economy, AS-AD, that simulates the periods of boom and bust that characterize modern economies.